

KOUCHIBOUGUAC, NB, Canada

WMO: 716700

Lat: 46.789N

Lon: 65.013W

Elev: 112

StdP: 14.64

Time Zone: -4.00 (NAA)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.0	(c) -3.6	(d) -15.9	(e) 2.3	(f) -5.9	(g) -11.4	(h) 3.0	(i) -0.9	(j) 14.5	(k) 22.4	(l) 14.0	(m) 20.8	(n) 3.3	(o) 260	(p) 0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	86.4	69.7	83.3	68.2	80.2	66.5	72.5	81.8	70.9	79.1	69.2	76.8	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	109.2	76.5	68.1	103.5	75.3	66.4	97.7	73.6	36.2	82.3	34.8	78.5	33.4	77.1	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.6	11.7	9.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-15.6	92.4	4.0	2.6	-18.5	94.3	-20.8	95.8	-23.1	97.3	-26.0	99.2
			-16.0	75.5	3.9	1.3	-18.8	76.4	-21.2	77.2	-23.4	77.9	-26.2	78.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	17.7	18.8	27.2	39.0	50.4	60.0	68.4	66.6	58.5	46.8	35.3	23.8		
	DBStd	19.49	11.52	10.32	9.35	7.21	7.73	7.15	5.79	5.29	7.36	7.38	9.13	10.05		
	HDD50	4445	1002	875	707	341	89	6	0	0	11	153	448	813		
	HDD65	8387	1467	1295	1171	781	459	183	33	45	218	566	891	1278		
	CDD50	1829	0	0	2	10	101	306	569	515	266	53	7	0		
	CDD65	296	0	0	0	0	6	33	137	95	24	1	0	0		
	CDH74	2637	0	0	3	7	122	388	1178	740	191	8	0	0		
	CDH80	704	0	0	0	0	30	107	355	169	43	0	0	0		
	WSAvg	3.7	4.1	4.5	4.6	4.0	3.7	3.4	3.0	2.9	3.1	3.3	4.1	4.2		
	PrecAvg	45.6	3.9	3.1	3.7	3.3	3.9	3.6	4.1	3.3	3.7	4.4	4.1	4.2		
(15) Precipitation	PrecMax	57.4	6.0	5.5	6.7	5.1	6.6	6.7	7.4	6.5	7.8	7.6	6.3	9.1		
	PrecMin	35.2	1.7	1.2	1.4	1.3	1.1	1.3	1.4	1.3	1.3	1.2	2.4	1.3		
	PrecStd	5.1	1.1	1.1	1.3	1.0	1.5	1.1	1.4	1.3	1.6	1.7	1.2	1.6		
	PrecStd	5.1	1.1	1.1	1.3	1.0	1.5	1.1	1.4	1.3	1.6	1.7	1.2	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	48.4	60.7	72.5	84.2	89.1	91.5	88.2	85.3	74.5	64.3	51.8		
		MCWB	49.1	45.6	47.9	54.8	65.0	67.5	71.6	70.9	70.3	64.3	62.4	50.5		
	2%	DB	43.0	41.8	49.5	62.5	77.1	83.0	87.0	84.4	79.0	67.2	58.7	45.3		
		MCWB	40.9	38.5	40.3	51.6	60.5	66.0	70.8	69.5	66.8	59.0	56.1	43.9		
	5%	DB	37.5	37.0	44.2	57.3	71.0	78.4	84.0	81.4	75.1	62.8	54.2	40.7		
		MCWB	35.8	33.7	38.3	48.4	57.7	63.8	69.1	67.6	65.0	57.9	50.8	39.4		
	10%	DB	33.4	33.1	40.1	52.3	65.2	74.2	80.8	78.3	71.3	59.1	49.5	36.5		
		MCWB	32.1	31.5	35.0	44.5	54.7	62.2	67.7	66.4	63.3	54.5	46.2	35.0		
	0.4%	WB	49.5	45.8	52.4	59.2	66.8	71.6	75.0	74.0	72.1	64.9	62.8	50.7		
		MCDB	50.3	47.6	60.1	67.5	77.8	81.4	85.7	82.9	81.7	69.9	64.5	51.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.7	39.0	43.0	54.4	62.8	68.9	73.1	71.8	69.6	61.7	56.8	44.3		
		MCDB	42.5	41.5	46.9	59.3	73.3	78.0	83.1	80.2	75.3	65.1	58.3	45.2		
	5%	WB	35.7	34.2	38.9	49.6	59.6	66.4	71.5	70.2	67.3	58.4	51.5	39.8		
		MCDB	37.3	35.8	43.8	55.7	66.8	74.7	80.2	77.9	72.8	62.1	53.4	40.8		
	10%	WB	32.4	31.9	35.5	45.3	56.2	63.7	69.7	68.4	64.7	54.9	46.2	35.4		
		MCDB	33.3	33.1	39.4	51.3	64.0	71.5	77.6	75.2	70.4	58.4	48.9	36.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	18.9	19.1	19.2	22.2	20.8	21.1	20.5	20.8	17.5	15.0	14.1		
		MCDBR	20.4	20.6	23.6	29.5	32.4	28.9	25.3	24.8	24.5	22.4	20.3	16.2		
		MCWBR	20.1	17.6	16.7	19.2	17.8	15.0	12.1	12.6	14.3	15.8	17.9	16.0		
		MCDBR	20.9	19.4	21.4	26.7	27.0	24.0	21.0	20.8	21.1	18.7	19.4	16.7		
	5% WB	MCWBR	20.8	17.6	16.7	19.3	17.0	14.0	11.3	11.9	14.0	15.0	18.7	16.8		
		MCWBR	20.8	17.6	16.7	19.3	17.0	14.0	11.3	11.9	14.0	15.0	18.7	16.8		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.282	0.319	0.350	0.368	0.388	0.413	0.391	0.357	0.327	0.302	0.282		
	taud		2.408	2.378	2.383	2.406	2.411	2.348	2.319	2.371	2.453	2.534	2.549	2.500		
	Ebn at Noon		258	285	288	287	285	278	269	271	272	265	252	248		
	Edh at Noon		22	28	32	35	36	38	39	36	30	24	19	18		
(44) All-Sky Solar Radiation	RadAvg		435	669	1056	1339	1590	1715	1748	1528	1209	758	418	318		
	RadStd		43	70	71	128	162	130	122	101	107	71	51	33		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page